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RESEARCH

Advances and challenges in AI-assisted MRI for lumbar disc degeneration detection and classification

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Abstract

Purpose Intervertebral disc degeneration (IDD) is a major contributor to chronic low back pain. Magnetic resonance imaging (MRI) serves as the gold standard for IDD assessment, yet manual grading is often subjective and inconsistent. With advances in artificial intelligence (AI), particularly deep learning, automated detection and classification of IDD from MRI has become increasingly feasible. This narrative review aims to provide a comprehensive overview of AI applications—especially machine learning and deep learning techniques—for MRI-based detection and grading of lumbar disc degeneration, highlighting their clinical value, current limitations, and future directions.

Methods Relevant studies were reviewed and summarized based on thematic structure. The review covers classical methods (e.g., support vector machines), deep learning models (e.g., CNNs, SpineNet, ResNet, U-Net), and hybrid approaches incorporating transformers and multitask learning. Technical details, model architectures, performance metrics, and representative datasets were synthesized and discussed.

Results AI systems have demonstrated promising performance in automatic IDD grading, in some cases matching or surpassing expert radiologists. CNN-based models showed high accuracy and reproducibility, while hybrid models further enhanced segmentation and classification tasks. However, challenges remain in generalizability, data imbalance, interpretability, and regulatory integration. Tools such as Grad-CAM and SHAP improve model transparency, while methods like few-shot learning and data augmentation can alleviate data limitations.

Conclusion AI-assisted analysis of MRI for lumbar disc degeneration offers significant potential to enhance diagnostic efficiency and consistency. While current models are encouraging, real-world clinical implementation requires further advancements in interpretability, data diversity, ethical standards, and large-scale validation.

Keywords Intervertebral disc degeneration · Artificial intelligence · Medical imaging · MRI · Diagnosis

Introduction

Intervertebral Disc Degeneration (IDD) is one of the leading causes of chronic low back pain [1–3]. Studies have shown that with increasing age, the structure of the intervertebral disc gradually degenerates, leading to a loss of normal biomechanical function, which is closely associated with the onset of low back pain [4–6]. Epidemiological data suggest that approximately 80% of adults will experience varying degrees of low back pain in their lifetime, with a significant proportion of cases being related to disc degeneration [7]. In addition to natural aging, IDD is influenced by genetic, lifestyle, and environmental factors, such as prolonged heavy lifting and poor posture [5, 8].

Magnetic Resonance Imaging (MRI) is widely recognized as the gold standard for diagnosing IDD because it provides non-invasive, detailed imaging of degenerative changes in the intervertebral disc [9]. MRI accurately assesses disc height, morphology, signal intensity, and Modic changes, enabling clinicians to determine the severity of degeneration [10]. Furthermore, MRI allows for the standardized grading of disc degeneration using the Pfirrmann classification system, which has become an essential tool for quantifying the degree of disc degeneration in clinical practice [11].

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While the Pfirrmann classification system is widely used to diagnose disc degeneration, manual grading relies on the subjective judgment of radiologists, which introduces variability in diagnostic outcomes [12]. Several studies have demonstrated significant differences in inter-observer agreement (i.e., consistency between different radiologists) and intra-observer agreement (i.e., consistency of the same radiologist over time) in grading disc degeneration [11]. This subjectivity not only reduces the reliability of diagnosis but also increases the time required for assessment, especially when dealing with large volumes of MRI data [11]. In addition, manual grading systems face limitations in integrating multimodal information, making it difficult to incorporate clinical and imaging features of patients comprehensively. These limitations further restrict the widespread application of manual grading in clinical settings [7].

In recent years, Artificial Intelligence (AI), particularly deep learning, has made significant strides in the field of medical imaging analysis [13]. Convolutional Neural Networks (CNN), a key AI technique, can automatically identify complex pathological patterns from imaging data by training on large datasets, thus providing accurate quantitative assessments and minimizing human subjectivity [14]. AI models can process large volumes of MRI data in a short period while maintaining high consistency across multiple assessments, thereby greatly improving the efficiency and accuracy of diagnoses [12]. AI technology has shown great potential in the automatic grading of disc degeneration. By training on extensive MRI datasets, AI models can automatically detect and grade disc degeneration with minimal human intervention, significantly improving diagnostic accuracy [7].

This review aims to summarize the current applications of Artificial Intelligence in the automatic detection and grading of lumbar intervertebral disc degeneration through MRI. By systematically reviewing existing studies, we will explore the advantages of AI in improving diagnostic accuracy and reducing human subjectivity. Additionally, we will discuss the future potential of AI in clinical practice, providing theoretical and technical support for the automated diagnosis and personalized treatment of IDD.

Literature search strategy

Two radiologists conducted a comprehensive literature search independently across major datasets of PubMed, Embase, and Web of Science (including Medline) to investigate existing research regarding artificial intelligence-assisted MRI diagnosis in lDD. The combination of following key terms were used for literature search: "Deep Learning or Network", "Artificial Intelligence or AI", "Machine learning", "Lumbar or Disc", "Degeneration or Degenerative" and "MRI", and references from included researches were reviewed to identify any relevant studies.

The two radiologists reviewed the retrieved literatures, subjectively selected publications deemed pivotal for artificial intelligence based diagnosis and treatment of IDD, and conducted a narrative synthesis. Ultimately, 17 articles from 2014 to 2024 were included in this work, as shown in Tables 1 and 2.

MRI in the diagnosis of IDD

MRI is the gold standard for diagnosing intervertebral disc degeneration due to its superior soft tissue contrast and ability to visualize the intricate structure of the spine without ionizing radiation. The most commonly used MRI sequences for evaluating IDD include T1-weighted, T2-weighted, and STIR (Short Tau Inversion Recovery) images, each providing unique insights into the state of the intervertebral discs and surrounding tissues [10]. T2-weighted images are particularly important in assessing the hydration status of the discs, with decreased signal intensity in T2 images correlating with a loss of water content, a hallmark of disc degeneration [7, 15].

In addition to detecting disc degeneration, MRI is also effective in identifying related pathologies such as Modic changes in the vertebral endplates, disc bulging, and herniation, all of which can contribute to low back pain [16]. High-intensity zones (HIZ) seen on T2-weighted images have also been linked to annular tears and discogenic pain, further demonstrating the versatility of MRI in capturing a wide range of degenerative changes [17]. Despite its advantages, MRI does have limitations, particularly in distinguishing between symptomatic and asymptomatic degenerative changes. Many individuals show radiographic evidence of degeneration without corresponding clinical symptoms, making it difficult to determine the clinical relevance of MRI findings [3]. As such, integrating clinical evaluation with imaging findings remains crucial for accurate diagnosis and treatment planning.

Current techniques applied for IDD grading

Manual classification systems

The Pfirrmann classification system is widely used for grading intervertebral disc degeneration (IDD) based on MRI findings. This system assesses changes in disc structure, including the loss of disc height, changes in signal intensity, and the presence of herniation or disc bulging, which are

Table 1 Application of AI technologies in lumbar disc degeneration grading

AI technology	Model type	Application area	Representative study	Advantages	Limitations
Support Vector Machine (SVM)	Traditional Machine Learning	Automated Detection and Classification of IDD	Oktay et al. (2014)	Accurate in processing manually extracted features, performs comparably to experts	Requires manual feature extraction, difficult to scale to large datasets
Decision Trees and Random Forests	Traditional Machine Learning	IDD Grading and Decision Trees	Early model applications	Simple and easy to use, suitable for small datasets	Sensitive to data noise, struggles with complex structures
Convolutional Neural Network (CNN)	Deep Learning	Automated MRI Detection and Grading	SpineNet (2017)	Learns image features automatically, no need for manual intervention, excellent performance	Highly dependent on data, lacks transparency ("black box" issue)
Hybrid Models (CNN + Transformer)	Deep Learning + Transformer	Precise Segmentation of Lumbar Structures and Degeneration Grading	SymTC (2024)	High spatial accuracy, combines attention mechanisms for better results	High computational complexity, long training times
Multi-task Learning Models	Deep Learning	Multi-task Processing: Segmentation, Classification, and Localization	Han et al. (2018)	Shares task features, improving overall performance and efficiency	Requires a large amount of high-quality annotated data
Reinforcement Learning (RL)	Reinforcement Learning	Adaptive Diagnostic Feedback	Gao et al. (2021)	Adjusts itself through feedback mechanisms, improving model performance	Limited application in medical imaging so far
Multimodal AI Models	Deep Learning with Multimodal Data	Comprehensive Analysis Combining MRI and Clinical Data	Future Trend	Integrates multimodal data, suitable for complex case analysis	Complex data integration, limited clinical validation

common features of degeneration [18]. It assigns scores on a scale of 1 to 5, with higher scores indicating more severe degeneration. Pfirrmann grading has been proven valuable in both clinical and research settings due to its simplicity and ability to standardize the evaluation of IDD [19]. However, despite its widespread use, the Pfirrmann system also possesses several limitations. One of the primary drawbacks is its reliance on subjective visual assessment, which can lead to significant inter- and intra-observer variability. Several studies have shown that different radiologists may interpret the same MRI images differently, resulting in inconsistent grading outcomes [11]. This inconsistency complicates the comparison of diagnostic results across studies or clinical practices. Moreover, the Pfirrmann system does not take into account other critical factors such as Modic changes or the presence of high-intensity zones, which have been shown to correlate with more advanced disc degeneration and low back pain [20]. These limitations highlight the need for more objective and comprehensive grading systems that can integrate additional imaging biomarkers beyond disc morphology. Some modifications to the Pfirrmann system have been proposed, but variability and the limited scope of these changes remain challenges in the consistent assessment of IDD [18].

Traditional machine learning approaches

Early machine learning models have shown promise in automating the grading of intervertebral disc degeneration (IDD) using MRI data. Algorithms such as Support Vector Machines (SVM), Decision Trees, and Random Forests have been applied to classify MRI features of IDD (Table 1). These models typically rely on handcrafted features, such as disc height, signal intensity, and disc shape, which are used to train the models to differentiate between normal and degenerated discs. Oktay et al. demonstrated the use of SVM for the automated classification of degenerative lumbar discs from MRI scans [21]. Apart from assessment of disc intensity and shape, they also incorporated texture features of the intervertebral discs into the construction of machine learning model, which includes information that cannot be directly captured by visual observation, and ultimately their model achieved an excellent accuracy of 92.81% [21]. Similarly, Beulah et al. extracted a hybrid of basic intensity, invariant moments, Gabor features from segmented intervertebral disc at T2-weighted sagittal MR image to construct SVM model for binary degenerative discs classification. Their model obtained an accuracy of 92.47% at validation dataset, and outperformed other machine learning models [22].

Studies have shown that machine learning models constructed based on features measured or calculated manually can achieve satisfactory efficacy in identifying degenerative disc disease. The advantage lies in the strong interpretability of the features used for model construction which are accepted by radiologists. Moreover, this method allows for easy acquisition of diagnostic results, making it convenient for clinical application. On the other hand, the disadvantage

Table 2 Overview of recent studies on AI-Driven detection and classification of lumbar disc degeneration using MRI

Study	Year	AI technique/model	Dataset	Key findings	Performance metrics	Clinical implications
Ghosh & Chaudhary (Computerized Medical Imaging) [28]	2014	Random Forest	212 MRI scans	High localization accuracy (98%) for disc detection and segmentation of disc and dural sac	Dice Similarity Index: 0.87 (dural sac), 0.84 (intervertebral disc)	First step toward CAD system for lower back issues based on MRI segmentation
Grob et al. (European Spine Journal) [29]	2022	SpineNet (Deep Learning)	882 MRI scans	SpineNet shows moderate to substantial agreement with radiologists for disc degeneration (κ=0.72)	Class Average Accuracy: 55% for PG, 56% for SL, 43% for CCS	Reliable tool for automated grading in large cohorts
McSweeney et al. (Spine Journal) [30]	2023	SpineNet (Deep Learning)	1331 MRI scans (Northern Finland Birth Cohort)	SpineNet matched human reliability in DD grading and Modic change detection	Balanced accuracy: 78% for DD, 86% for MC; κ=0.68 (DD), κ=0.74 (MC)	Demonstrates robust performance across different datasets
Jamaludin et al. (Eur Spine J) [27]	2017	CNN (Deep Learning)	12,018 MRI scans from 2009 patients	CNN automated Pfirrmann grading with 95.6% accuracy in disc detection and labeling	Accuracy: 95.6%; κ>0.9 for inter-rater reliability	Comparable to radiologist accuracy in grading and segmentation
Han et al. (Neuroinformatics) [31]	2018	Deep Multiscale Multitask Learning	200 T1/T2-weighted MRI scans	High accuracy in localizing and grading multiple spinal organs	Mean Average Precision: 0.845	Facilitates diagnosis of lumbar neural foraminal stenosis and aids clinical workflow
Hai et al. (Computers in Biology and Medicine) [32]	2024	Semi-supervised learning (SeCoFixMatch)	Unspecified MRI dataset	Outperforms baseline model with 40 labeled images vs. 200 labels, reducing annotation by 80%	Significant reduction in annotation effort	Enhances LDH diagnosis with minimal human annotation
Zheng et al. (Nature Communications) [25]	2022	Semantic Segmentation (BianqueNet)	Large population MRI dataset	High-precision segmentation for lumbar IVDD with excellent agreement with manual measurements	Mean Absolute Error: 0.08 Pfirmann grade; r=0.96	Automated system could assist in clinical practice and large-scale trials
Natalia et al. (PLOS ONE) [33]	2024	ResNet-50 and Ensemble Decision Trees	515 MRI studies	Achieved 88.1% accuracy in Pfirrmann grade prediction for lumbar spine MRI	Mean Accuracy: 88.1%	Addresses clinical workforce limitations with automated MRI annotation
Ma et al. (J Appl Clin Med Phys) [34]	2020	Faster R-CNN (Deep Learning)	1,500 MRI scans (cervical spine)	Faster R-CNN achieved high accuracy in detecting cervical spinal cord lesions	Mean Average Precision: 88.6% (ResNet-50), 72.3% (VGG-16)	Potential aid for radiologists in detecting spinal cord lesions
Niemeyer et al. (Investigative Radiology) [12]	2021	CNN (Deep Learning)	7,948 intervertebral discs	CNN outperformed human reliability in Pfirrmann grading (κ=0.92)	Sensitivity: 90.2%; Precision: 92.5%	Reliable for use in longitudinal clinical studies

is that traditional machine learning methods do not eliminate the influence of subjective bias which may struggle with variability in disc morphology and signal intensity across different patients and face limitations in handling large-scale, high-dimensional MRI data, which often necessitates feature engineering.

Deep learning approaches

Deep learning, a subset of machine learning, which is based on a neural network structure inspired by the human brain [23, 24], especially Convolutional Neural Networks (CNNs), has revolutionized in medical imaging analysis, providing a significant improvement in the automatic detection and grading of IDD. CNN consists of multiple types of layers, including convolutional layers, pooling layer, activation layer, and fully connected layer. As the layers get deeper, more and more abstract features are extracted form images, making these features difficult to interpret. Owing to the architecture, CNNs can learn hierarchical and abstract features directly from the raw image data, eliminating the need for manual feature extraction, which was a major limitation of earlier machine learning models.

Several studies have utilized deep learning for degenerative disc disease grading. Niemeye et al. [12] propose a classifier for disc degeneration based on a deep convolutional neural network that trained on a large, manually evaluated data set of 1599 patients with 7948 intervertebral discs. Their system achieved highly reproducible grading with superior performance with an average sensitivity of more than 90%.

Their work effectively addressed the limitations associated with small sample sizes, yet it was also confronted with the challenge of imbalanced training samples. Zheng et al. [25] utilized a deep learning network based on T2-weighted images to achieve quantitative assessment of IDD, thereby enhancing the reproducibility and efficiency of IDD diagnosis compared to traditional subjective evaluation methods. One of the pronounced advantages of deep learning technology is that it simplifies the diagnostic process through end-to-end learning and provides entirely objective results and exhibits advantage of high reproducibility, overcoming the limitations of traditional machine learning methods and manual assessment by radiologists.

The relatively high variability within classes and small differences between classes in MR images present challenges for classification algorithms in automatically grading degenerated IDD. Improving deep learning algorithms provides assistance in the grading of IDD. Gao et al. introduced a push-pull regularization (PPR) network based on deep learning that improved the performance of IDD grading across multiple datasets [7]. Similarly, Rudisill et al. applied deep learning models to predict adjacent segment degeneration following spinal surgery, demonstrating high accuracy and reliability in clinical applications [26]. CNN-based models like ResNet and U-Net have been particularly effective in segmenting and classifying spinal structures, including intervertebral discs and vertebrae. The core idea of ResNet is to design each layer of the network to learn the residual between the input and output, rather than directly learning the mapping relationship. This structure allows gradients to propagate directly during training, thereby alleviating the vanishing gradient problem. As a result, it is possible to construct very deep networks while maintaining good performance. The key feature of U-Net is the skip connection, which directly connects the feature maps from the encoder to the corresponding layers in the decoder. Through these skip connections, U-Net can fully utilize high-resolution features to achieve high-precision segmentation. For instance, the SpineNet model developed by Jamaludin et al. (2017) achieved high accuracy in automatically grading disc degeneration, matching the performance of expert radiologists [27]. These models can handle complex image features, such as Modic changes and high-intensity zones, making them more robust in real-world clinical applications (Table 2). Figure 1 provides a detailed explanation of the feature extraction processes and feature details of traditional machine learning models and deep learning models, as well as the workflow for identifying intervertebral disc degeneration using AI tools.

Hybrid models and advanced architectures

Recent advancements in artificial intelligence have led to the development of hybrid models that combine CNNs with other AI techniques, such as transformers and reinforcement learning, to further enhance the accuracy and robustness of IDD detection and grading. Chen et al. proposed a hybrid SymTC model that combines the strengths of both CNNs and Transformers for the instance segmentation of lumbar MRIs [35]. This model leverages the spatial precision of CNNs with the self-attention mechanism of transformers, allowing for more accurate segmentation of vertebral structures and intervertebral discs. Additionally, multitask learning models, like the one developed by Han et al., integrate multiple tasks such as segmentation, classification, and localization into a single framework. These models improve the overall performance by sharing learned features across tasks, which is particularly beneficial when working with MRI data that contains both anatomical and pathological information [31]. Such hybrid approaches not only enhance

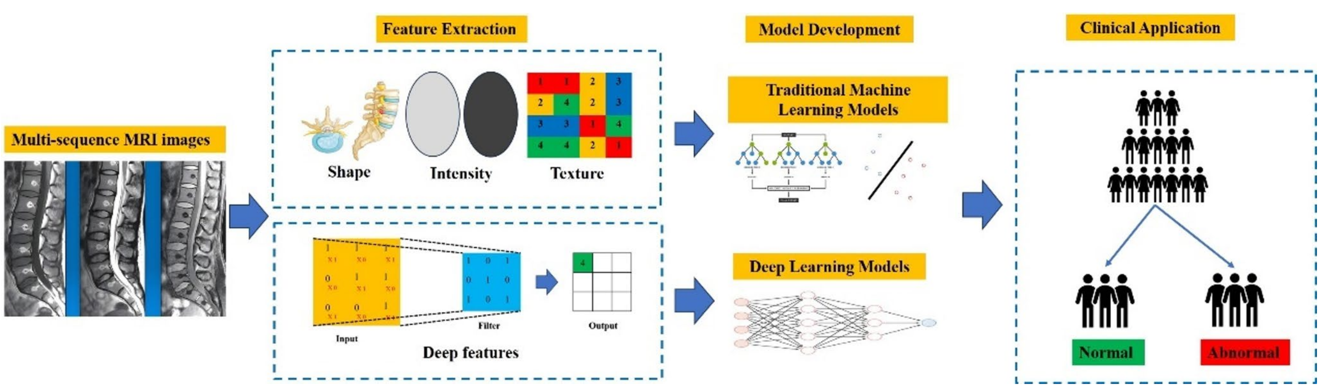

Fig. 1 Workflow of AI-assisted identification and grading of intervertebral disc degeneration from lumbar MRI images. The process begins with multi-sequence MRI image acquisition, including T1-, T2-, and STIR-weighted scans. Features are then extracted through traditional handcrafted methods that capture disc shape, intensity, and texture, or via deep learning approaches that automatically learn hierarchical features from raw images. Model development uses either traditional machine learning algorithms (e.g., decision trees, support vector machines) or deep learning networks (e.g., convolutional neural networks) to classify disc status. Finally, the models output predictions distinguishing normal from abnormal discs, providing decision support for radiologists and facilitating clinical diagnosis of IDD

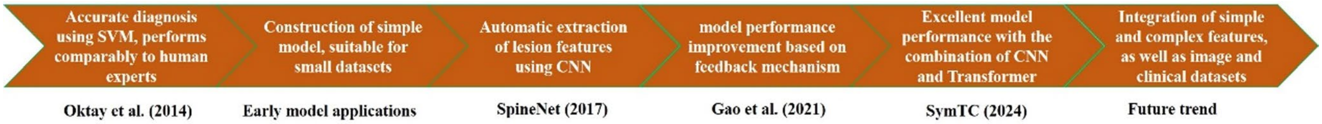

Fig. 2 Timeline of key developments in AI technologies for automatic detection and grading of intervertebral disc degeneration (IDD) from MRI. This figure summarizes major milestones from early applications using traditional machine learning algorithms, such as support vector machines (SVM) with manual feature extraction [21], to automatic feature learning with convolutional neural networks (SpineNet, 2017), advancements in model performance via feedback mechanisms [7], and recent hybrid models combining CNN and Transformer architectures (SymTC, 2024). Future trends point toward integrating simple and complex features with multimodal clinical and imaging datasets for more robust and comprehensive IDD assessment

diagnostic accuracy but also reduce the computational load, making them more suitable for clinical use.

A timeline highlighting the key milestones in AI-assisted IDD detection is illustrated in Fig. 2.

Clinical utility and workflow integration

The clinical application of AI models is expanding, and integrating AI into existing clinical workflows is a key challenge. AI models can provide real-time automated analysis for radiologists, significantly reducing the time required for image evaluation [36]. However, for AI to be fully integrated into clinical settings, it needs to ensure ease of use, compatibility with existing Picture Archiving and Communication Systems (PACS), and reliable clinical decision support [13]. Compared to expert radiologists, AI models can provide consistent grading results more efficiently when analyzing large-scale MRI data. Studies have shown that deep learning models perform similarly to radiologists in grading disc degeneration and, in some cases, even surpass human experts [14]. For instance, models like SpineNet have demonstrated high accuracy in automatic grading, especially when evaluating Modic changes and high-intensity zones [27].

For the diagnosis of IDD, the diagnostic results of AI tools provide radiologists with a second reference in addition to their own experience-based judgment, mitigating elementary errors in daily workflows. When the performance of AI tools is sufficiently excellent, their role may shift from being an assistant to a trusted partner, significantly reducing the workload of radiologists and improving work efficiency. However, it should be noted that the results of AI tools need to be verified by radiologists. So far, they do not have the conditions and capabilities to conduct independent diagnoses, nor do they have acquired such confidence from clinic workers.

Challenges and solutions

Data quantity and quality

The performance of AI models largely depends on the quality and diversity of the training data. Limited and imbalanced data distribution can both affect the performance of the model, leading to weak generalization ability in different clinic scenarios [15]. Currently, there are some methods that have shown certain effects in alleviating the issue of data scarcity in artificial intelligence research. Few-shot learning optimizes learning algorithms and strategies to enable models to learn effectively from a limited number of samples [37, 38]. Transfer learning leverages pre-trained models as a starting point, transferring the knowledge acquired to new tasks to reduce the risk of overfitting and address the issue of limited data [39, 40]. Data augmentation generates new training samples by applying operations such as rotation, translation, flipping, and scaling to images, thereby enhancing the model's generalization ability [41–43]. Meanwhile, image generation algorithms based on Generative Adversarial Networks (GANs) and diffusion models also show potential in addressing data scarcity. GANs employ adversarial training between a generator and a discriminator. The generator progressively learns to create realistic images, while the discriminator attempts to distinguish between real and fake images. Diffusion networks, on the other hand, gradually add noise to the data to diffuse it into Gaussian noise, and then reverse this process to restore the original data and generate images [44–48]. Meanwhile, constructing high-quality datasets of IDD cases from multiple centers, with various devices and diverse patient populations, is also key to improving the robustness of AI models for IDD diagnosis.

Explainability and transparency of AI models

In traditional machine learning models, models are trained based on features manually defined and extracted. As a result, radiologists are able to comprehend the basis for the model's decision-making, which means the interpretability is relatively high. However, it appears that the majority of current research on IDD diagnosis is now focusing on deep learning. Deep learning models, are often perceived as "black boxes," making their decision-making process difficult to explain [49–51]. In medical applications, this lack of transparency can reduce trust in AI diagnostic results by both clinicians and patients [14]. Therefore, improving the interpretability of AI models, allowing doctors to understand the basis of the model's decisions, is an important step toward the clinical application of AI.

Some methods have been proven to be effective in the interpretability analysis of deep learning. The most commonly used approach of Explainable AI in medical image analysis is visualization techniques, also called saliency mapping. Methods such as Gradient-weighted Class Activation Mapping (Grad-CAM), layer-wise relevance propagation (LRP) and SHapley Additive exPlanation(SHAP)are used to generate saliency maps which highlight the regions in the input image that are most relevant to the model's predictions [52, 53]. Xu et al. employed the SHAP method to interpret the significance of the spinal morphological parameters in predicting biological age [54]. Attention mechanisms can also help the model focus on specific regions of the image that are most important for making a decision and thus enhancing its interpretability [55, 56]. In the medical field, both doctors and patients need to have sufficient trust in diagnostic outcomes. If the decision-making process of a deep learning model is opaque, it may lead to clinical doctors being skeptical of the results, thereby impeding its application in real-world medical scenarios. Through interpretability analysis, doctors can better understand the basis for the model's decisions, thereby enhancing their confidence in the model. Patients are also more likely to accept diagnostic results that come with clear explanations. Interpretability analysis can help patients understand why the model has reached a particular diagnosis, thus increasing their acceptance of new technologies.

Ethical and regulatory considerations

As AI is increasingly applied in medicine, discussions about its ethical and regulatory challenges are growing. The safety and reliability of AI systems are among the key concerns of medical regulatory agencies. Deep learning models may inherit biases present in the training data, which can lead to inequitable diagnosis and treatment for certain patient populations [57, 58]. For instance, if a model is primarily trained on data from a specific demographic group, it may exhibit biases when diagnosing individuals from other groups. This not only affects the health of individual patients but can also exacerbate inequalities in the allocation of healthcare resources. We take that this issue is still caused by the category of insufficient data volume. Constructing multicenter, large-scale databases or employing various data augmentation algorithms to increase the quantity and balance of training samples for deep learning models can effectively address this problem. Additionally, AI technologies face challenges in protecting patient data privacy, particularly during large-scale data collection and model training [26]. To ensure the proper use of AI, strict regulatory frameworks and ethical guidelines must be established.

Future directions

With the continuous advancement of AI technologies, emerging methods such as reinforcement learning, attention mechanisms, and unsupervised learning offer promising prospects for IDD detection. Reinforcement learning can adjust itself through feedback mechanisms, while attention mechanisms enhance the capture of key features in MRI images, improving model accuracy [35]. Algorithmic improvements will continue to contribute to enhancing the accuracy of IDD diagnosis. Multimodal data fusion strategies are also considered to be promising. By combining MRI with other imaging modalities (e.g., CT, X-ray) and clinical data, AI models can further enhance their performance. Multimodal data provide more comprehensive diagnostic information and improve the model's ability to handle complex cases [31]. Autonomous decision-making AI agents in the medical field are anticipated and are on the path of development [59, 60], but there are still some preparatory works that need to be completed at present. AI's role in personalized medicine is becoming increasingly prominent, particularly in predicting patient treatment outcomes and providing individualized interventions. By analyzing large amounts of patient imaging and clinical data, AI can offer personalized diagnostic and treatment recommendations, thereby improving patient outcomes [4]. The real-world application of AI technology requires large-scale prospective clinical trials to validate its effectiveness. Trials assessing the performance of AI in real-world settings, including its application in different hospitals and patient populations [27] is needed and hold significant implications for their eventual clinical application.

Conclusion

This review summarizes the application of artificial intelligence in the automatic detection and grading of intervertebral disc degeneration (IDD) using MRI images. AI technology has significantly improved diagnostic accuracy and consistency, reducing the subjectivity of manual grading. While AI shows great potential in IDD detection, future efforts are needed to address challenges related to data diversity, model interpretability, and clinical implementation.

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Declarations

Competing interests The authors declare no competing interests.

Ethics approval This article does not contain any studies with human participants or animals performed by any of the authors, and therefore ethics committee approval is not applicable.

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